pmsimstats: Status Report on N-of-1 Trial Simulation Framework

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1 Background and Objective

The pmsimstats framework provides Monte Carlo simulation-based power analysis for aggregated N-of-1 clinical trial designs. It compares four trial designs – Open-Label (OL), Crossover (CO), Hybrid N-of-1, and Open-Label with Blinded Drug Cessation (OL+BDC) – on statistical power to detect a biomarker-treatment interaction. The motivating clinical application is the use of resting blood pressure as a predictive biomarker for prazosin response in PTSD, based on the Raskind et al. (2013) RCT analyzed in Murray et al. (in preparation).

The original simulation code was published in Hendrickson et al. (2020), *Frontiers in Digital Health* 2:13. The current work represents a comprehensive audit, revision, and consolidation of the codebase into a single reproducible research repository (`pmsimstats-ng`) housed as a zzcollab Docker workspace.

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2 Audit and DGP Revisions

A technical audit of the original publication code identified seven substantive corrections to the data-generating process (DGP), each justified by either statistical theory or clinical plausibility.

2.1 AR(1) autocorrelation (was compound symmetry)

The original code used compound symmetry (constant correlation ρ between all timepoint pairs). The revised code implements AR(1) decay: $\mathrm{Cor}(X_{t_i}, X_{t_j}) = \rho^{|t_i - t_j|}$. This change is both more clinically realistic (nearby measurements are more correlated than distant ones) and numerically beneficial: it expands the positive-definite-feasible region of the biomarker correlation parameter c_{bm} from a maximum of 0.25 to 0.49+.

2.2 Exponential BM-BR correlation decay

During off-drug periods, the correlation between the biomarker and the biological response should decay as the drug effect wanes. The original code used a step function (correlation drops to zero instantly at discontinuation). The revised code implements exponential decay tied to the pharmacokinetic half-life: $\mathrm{Cor}(bm, br_t) = c_{bm} \cdot e^{-\lambda_{cor} \cdot t_{sd}}$, where $\lambda_{cor} = \ln(2)/t_{1/2}$. This derivation is presented in the correlation decay derivation white paper.

2.3 nlme::lme with corCAR1 analysis model

The original analysis model used `lme4::lmer` with a random intercept only. Under the AR(1) DGP, this misspecified model absorbs residual autocorrelation into the error term, inflating Type I error to 13–17% for CO and OL designs. The revised code uses `nlme::lme` with `corCAR1(form = ~t|ptID)`, which models the continuous-time AR(1) residual structure directly. This corrects Type I error to the nominal 5% across all designs. The analysis model white paper provides the full derivation and validation.

2.4 Carryover scale factor removal

The original code applied a scale factor of 2 to the carryover half-life parameter, effectively halving the half-life without documentation. The parameter `carryover_t1half` should mean what it says: the time for the carryover effect to decay to half its initial value. The scale factor has been removed.

2.5 Timepoint-1 correlation fix

The original code guarded the BM-BR correlation assignment with `if (p > 1)`, skipping the first timepoint. This was a coding artifact to avoid an index error in the proportional scaling formula. Participants are on drug for 2.5–4 weeks at timepoint 1 and should have a biomarker-response correlation. The guard has been removed.

2.6 Hendrickson-aligned Gompertz function

The Gompertz growth function $f(t) = M \cdot e^{-d \cdot e^{-r \cdot t}}$ does not pass through the origin ($f(0) \neq 0$). The original publication code applied an offset-rescaling to guarantee $f(0) = 0$ and preserve the asymptote at M. The 2025 tidyverse rewrite inadvertently used a different formula that does not pass through the origin. This has been corrected: both pipelines now use the Hendrickson-aligned formula.

2.7 Updated default parameters

Autocorrelation base rate reduced from 0.8 to 0.7 (consistent with AR(1) decay). Biomarker correlation c_{bm} constrained to $\{0, 0.25, 0.50\}$ (maximum PD-feasible value under AR(1) is 0.49

for OL). Carryover half-life range updated from $\{0, 0.1, 0.2\}$ weeks (too short to produce genuine carryover effects) to $\{0, 0.5, 1.0\}$ weeks.

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3 Resolving the Positive Definiteness Problem

3.1 The problem

The DGP constructs a 26×26 covariance matrix Σ from user-specified correlation parameters and passes it to an MVN draw. When the resulting matrix is not positive definite (PD), the code silently projects it to the nearest PD matrix via `corpcor::make.positive.definite()`. Instrumentation of the original publication code revealed that this is not a rare edge case: 24.7% of covariance matrices required silent correction at the publication commit, rising to 63.0% after subsequent edits. All 162 matrices in the standard parameter sweep were ill-conditioned ($\kappa > 100$).

3.2 Root causes

The failures are concentrated in the OL+BDC design (83–92% failure rate) due to an abrupt expectancy-driven change in the ER (expectancy response) variance mid-trial. The OL+BDC design transitions from full expectancy ($e = 1$) during the open-label phase to half expectancy ($e = 0.5$) during the blinded drug cessation phase. Because the ER standard deviation is scaled by e, this creates a factor-of-two variance discontinuity within the same correlation block, violating the implicit assumption that the specified correlations are consistent with the marginal variances. The compound symmetry autocorrelation structure compounds the problem: it requires all same-factor timepoint pairs to have the same correlation regardless of their temporal separation, which becomes untenable when variances change mid-trial.

The OL design never fails because all timepoints have the same expectancy ($e = 1$), so there is no variance heterogeneity.

3.3 Resolution

The revised code resolves the PD problem through two mechanisms:

1. **AR(1) autocorrelation** replaces compound symmetry. Because AR(1) correlations decay with temporal distance ($\rho^{|t_i - t_j|}$), the correlation matrix has smaller off-diagonal entries for distant timepoints, making it substantially easier to satisfy the PD constraint. The maximum feasible c_{bm} increases from 0.25 to 0.49+.
2. **Pre-validation with constrained parameters.** The `validateParameterGrid()` function tests all parameter combinations for PD before simulation and reports failures upfront. The default parameter grid ($c_{bm} \leq 0.45$, $\rho = 0.7$) produces zero PD failures across all 162 combinations for all four trial designs.

The net result: the revised code runs the complete parameter sweep with zero silent corrections, eliminating the uncontrolled perturbation that confounded cross-design power comparisons in the original publication.

4 Resolving the Power Dropoff Problem

4.1 The problem

When carryover is introduced into the DGP (non-zero $t_{1/2}$), the statistical power to detect the biomarker-treatment interaction should decline because the carryover effect blurs the distinction between on-drug and off-drug periods. However, the magnitude and pattern of the power decline depends on the entire variable chain from trial design specification through the analysis model interaction term.

Five derived variables form this chain:

1. `ondrug` – binary path assignment (design-determined)
2. `tod` – cumulative time on drug (Gompertz input)
3. `tsd` – time since discontinuation (carryover input)
4. `Dbc` – continuous drug indicator in the analysis model
5. `bm:Dbc` or `bm:t` – the interaction coefficient tested

Carryover enters at step 3 and propagates through steps 4 and 5 to reduce the variance of the interaction term, inflating its standard error and reducing power.

4.2 The mechanism

For designs with both on-drug and off-drug periods (CO, Hybrid, OL+BDC), the analysis model uses `Dbc` as the treatment indicator. On drug, $Dbc = 1$. Off drug, $Dbc = (1/2)^{s \cdot tsd/t_{1/2}}$, where s is the scale factor and $t_{1/2}$ is the carryover half-life.

Without carryover ($t_{1/2} = 0$), Dbc is effectively binary (1 vs 0), maximizing the contrast between treatment conditions. With carryover, Dbc during off-drug periods is strictly positive (decaying from 1 toward 0), compressing the treatment contrast. The interaction coefficient `bm:Dbc` depends on the variance of Dbc, which decreases as carryover increases, inflating the standard error.

For the open-label design (no off-drug periods), the analysis model uses `bm:t` instead of `bm:Dbc`, so carryover has no effect on power – there are no off-drug periods for carryover to operate on.

4.3 Resolution

The revised code addresses the power dropoff problem at two levels:

1. **Correct DGP-analysis alignment.** The AR(1) autocorrelation in the DGP is matched by `corCAR1` in the analysis model. This prevents residual autocorrelation from being absorbed into the error term, which would artificially inflate or deflate power estimates.
2. **Clinically realistic carryover parameters.** The original half-life values (0.1–0.2 weeks) produce negligible carryover effects that decay to essentially zero within one inter-visit interval. The revised values (0.5–1.0 weeks) produce genuine carryover that persists across multiple

off-drug timepoints, enabling meaningful power comparisons between designs that differ in their susceptibility to carryover.

3. **Conditional analysis formula.** The analysis model adapts to the design: `bm:Dbc` when there is within-subject treatment variation (CO, Hybrid, OL+BDC), and `bm:t` when there is not (OL). This prevents the open-label power estimate from being confounded by a degenerate treatment indicator.

The 1,000-replicate validation confirms that Type I error rates are within 2.5–5.8% across all designs and carryover conditions, and power estimates decrease monotonically with increasing carryover half-life as expected.

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5 Consolidated Repository

The pmsimstats-ng repository consolidates the original publication code (`orig`, data.table) and the 2025 tidyverse rewrite (`2025`) into a single zzcollab Docker workspace with shared data, documentation, and test infrastructure.

5.1 Architecture

The two simulation pipelines coexist independently:

- **orig** (`R/`, 1973 lines, 10 files): data.table-based package functions loaded via `devtools::load_all(".")`. Uses `buildSigma()`, `generateData()`, `lme_analysis()`, `generateSimulatedResults()`. This is the gold standard.
- **2025** (`analysis/2025/01-pm-functions.R`, 1023 lines): tidyverse-based monolithic file loaded via `source()`. Uses `build_sigma_matrix()`, `generate_data()`, `lme_analysis()`, `generate_simulated_results()`. Validated bit-identical to orig for the same inputs.
- **simple** (`simple/simulation.R`, 577 lines): self-contained pedagogical simulation using direct mean moderation (not MVN correlation) and OLS on phase means (not LME). Serves a different purpose and is architecturally independent.

5.2 Validation

- 67 unit tests covering `cumulative()`, `modgompertz()`, `buildtrialdesign()` (OL, CO, Hybrid, tpb), `buildSigma()` (PD, dimensions, labels, AR(1), lambda_cor, Cholesky), `generateData()` (output shape, caching, outcome identity), and `lme_analysis()` (OL, CO, p-value bounds).
- 3-level alignment test: Gompertz, sigma matrix, and generated data are bit-identical between orig and 2025 for the same seed.
- A bug in `lme_analysis()` (uninitialized `op` list when called without arguments) was discovered and fixed through the test suite.

5.3 Reproducibility infrastructure

- Docker image via zzcollab `analysis` profile (`rocker/tidyverse:4.5` + renv)
- `renv.lock` capturing all dependencies (nlme, data.table, furrr, corpcor, MASS, ggplot2)
- `Makefile` with targets: `make r` (container shell), `make rstudio`, `make check-renv`, `make docker-build`, `make test`
- GitHub repository: `rgt47/pmsimstats-ng`

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6 Collaborator Deliverables

6.1 Goal 1: Did prazosin improve PTSD symptoms?

Implemented as `analyze_trial_extended()` in `R/carryover_analysis.R`. Fits the LME model to actual trial data and extracts both the main drug effect (`Dbc` coefficient) and the biomarker-treatment interaction (`bm:Dbc`), with 95% confidence intervals, variance components, predicted response curves per participant, and a clinical decision threshold (the blood pressure value at which predicted benefit exceeds a specified CAPS improvement). Status: complete.

6.2 Goal 2a: Characterize carryover in actual data

Implemented as `characterize_carryover()` in `R/carryover_analysis.R`. Sweeps a range of candidate carryover half-lives (default 0.25 to 8 weeks), fits the LME model at each half-life with exponentially decaying `Dbc`, compares model fit via AIC/BIC and log-likelihood, and identifies the best-fitting half-life. Validated on simulated data: correctly recovers the true $t_{1/2} = 1.0$ from data generated with that half-life. Status: complete.

6.3 Goal 2b: Identify best sequence of response measures

Conceptual design only. Would require per-timepoint residual analysis to identify where the carryover signal is strongest in the observed data. The existing model infrastructure (`full_model_out = TRUE`) provides all necessary outputs (fitted values, residuals, random effects per participant). Status: not yet implemented.

6.4 Goal 3a: Simulation power calculations

Complete. The revised simulation produces 1,000-replicate power estimates with proper Type I error control (2.5–5.8% across all designs), zero PD failures across all 162 parameter combinations, sigma matrix caching for performance (8 minutes for 200 reps, 47 minutes for 1,000 reps), and parallel execution via furrr. Status: complete.

6.5 Goal 3b: Actual analysis output structure

Implemented in `analyze_trial_extended()`. Returns the full model object, predicted treatment effect as a continuous function of biomarker value, per-participant predicted response curves, and

clinical decision thresholds. The function can be applied directly to the bundled `CTdata.rda` dataset from the Raskind et al. trial. Status: complete.

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7 White Paper Documentation

The consolidated repository includes seven technical white papers documenting the audit findings, theoretical derivations, and validation results.

7.1 Revised Power Analysis (04-revised-power-analysis.pdf)

A 13-page comprehensive summary of all changes from v0.1.0 to v0.2.0, written for the original authors. Documents each DGP revision with before/after code, parameter tables, and heatmap comparisons. Includes the validation results showing Type I error correction and the complete parameter grid with power estimates at 200 and 1,000 replicates. Serves as the primary reference for understanding what changed and why.

7.2 AR(1) Residual Correlation (06-ar1-residual-correlation.pdf)

Derives why the original `lme4::lmer` random-intercept model produces inflated Type I error (13–17%) under the AR(1) DGP and validates that `nlme::lme` with `corCAR1` corrects this to the nominal 5%. Presents a simulation study comparing the two analysis models across 1,000 replicates under the null ($c_{bm} = 0$) for all four trial designs. Shows that the random-intercept model absorbs the AR(1) residual autocorrelation into a biased variance estimate, shrinking standard errors and inflating test statistics.

7.3 Biomarker Correlation Decay (08-biomarker-correlation-decay.pdf)

Derives the exponential decay formula for the biomarker-response correlation during off-drug periods from pharmacokinetic first principles. When each participant's drug effect decays by the same fraction per unit time ($e^{-\lambda t}$, where $\lambda = \ln 2/t_{1/2}$), the between-participant spread of drug effects decays proportionally, preserving the coefficient of variation. This implies the correlation between biomarker and drug-mediated response decays at the same rate: $\mathrm{Cor}(bm, br_t) = c_{bm} \cdot e^{-\lambda t_{sd}}$. The derivation shows this is the unique decay function that maintains the proportional relationship between individual drug effects and the biomarker.

7.4 Positive Definiteness Failures (07-positive-definiteness-failures.pdf)

Presents the instrumented measurement of positive definiteness failures across four code versions and 162 parameter combinations. Documents that 24.7% of covariance matrices fail PD at the publication commit, rising to 63.0% after subsequent edits, with all matrices ill-conditioned ($\kappa > 100$). Traces the root cause to the OL+BDC design's abrupt expectancy-driven variance change (83–92% failure rate) and the compound symmetry autocorrelation structure. Evaluates the perturbation introduced by `make.positive.definite()` and its consequences for cross-design power comparisons. Recommends AR(1) autocorrelation, pre-validation, and parameter constraints as solutions.

7.5 Carryover Correlation Artifact (09-carryover-correlation-artifact.pdf)

Analyzes the BM-BR correlation mechanism in the Hendrickson DGP. The biomarker-treatment interaction emerges from differential correlation in the MVN draw: when on drug, $\mathrm{Cor}(bm, br) = c_{bm}$; when off drug, the correlation decays or drops to zero. This means participants with high biomarker values tend to have large drug effects (high BR means) and participants with low biomarker values tend to have small drug effects, creating the interaction detectable by the analysis model. The paper traces how this correlation propagates through the sigma matrix to produce the observed power pattern across designs and parameter values.

7.6 Response Parameter Sensitivity (18-response-parameter-sensitivity.pdf)

Compares Figure 5 (response parameter sensitivity analysis) between the original publication code and the revised code. Figure 5 sweeps the Gompertz response parameters (max, disp, rate) and standard deviations across a grid to assess which parameters most strongly influence power. Documents the differences introduced by the DGP revisions and validates that the qualitative conclusions about parameter sensitivity are robust to the corrections.

7.7 Revision Summary (05-revision-summary.pdf)

Traces the impact of each individual commit (from the publication code through the Ron Thomas edits to the current HEAD) on the power estimates. Reproduces the Figure 4 heatmap at each commit to show how each change affected the power surface. Documents which changes improved power estimates (AR(1) expanding the feasible region), which degraded them (proportional BM-BR scaling reducing effective correlation), and which were neutral (code cleanup without DGP changes).

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8 Presentation Outline (7 Slides)

8.1 Slide 1: Problem and Motivation

- N-of-1 trial designs for biomarker-treatment interactions
- Four designs compared: OL, CO, Hybrid, OL+BDC
- Clinical context: prazosin for PTSD, blood pressure as predictive biomarker (Raskind et al. 2013, Hendrickson et al. 2020)
- Question: which trial design has the best power to detect who responds?

8.2 Slide 2: Audit Findings – Seven DGP Corrections

- AR(1) autocorrelation (was compound symmetry)
- Exponential BM-BR correlation decay (was step function)
- nlme + corCAR1 analysis model (was lmer, Type I error 13–17% corrected to 5%)
- Scale factor removal, timepoint-1 fix, Gompertz alignment
- Updated defaults: rho=0.7, c.bm $\leq$ 0.45, t_half 0–1.0 wks

8.3 Slide 3: Resolving the PD Problem

- Original: 24.7–63.0% of covariance matrices silently corrected
- Root cause: OL+BDC expectancy-driven variance discontinuity under compound symmetry
- Solution: AR(1) + parameter constraints = zero PD failures across all 162 combinations
- Consequence: cross-design power comparisons are now unconfounded

8.4 Slide 4: Resolving the Power Dropoff Problem

- Carryover erodes on/off-drug contrast through the variable chain: ondrug → tod → tsd → Dbc → bm:Dbc
- Original half-lives (0.1–0.2 wks) too short for genuine carryover
- Revised half-lives (0.5–1.0 wks) produce clinically meaningful power differences across designs
- Conditional formula: bm:Dbc (crossover) vs bm:t (open-label)

8.5 Slide 5: Consolidated Framework

- Two validated pipelines: data.table (orig, gold standard) + tidyverse (2025, bit-identical)
- 67 unit tests, Docker reproducibility, zzcollab workspace
- 1,023-line tidyverse pipeline (38% reduction from cleanup)
- GitHub: rgt47/pmsimstats-ng

8.6 Slide 6: Collaborator Deliverables

- analyze_trial_extended(): drug effect + interaction + predicted curves + clinical thresholds
- characterize_carryover(): sweep half-lives, compare AIC, find best fit
- 1,000-rep validated power with Type I error 2.5–5.8%
- 7 white papers documenting methodology

8.7 Slide 7: Next Steps

- Goal 2b: per-timepoint residual analysis for carryover signal localization
- Final figure reproduction with consolidated code and revised defaults
- Manuscript describing the revised methodology and consolidated results
- Sensitivity analysis: carryover decay shape (exponential vs Weibull), separate correlation decay rate

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